

TASK 2: SENSOR-BASED SIMULATION

AUTOMATIC STREET LIGHT USING LDR SENSOR

1. AIM

To design and simulate an automatic street light system using an LDR (Light Dependent Resistor) sensor and Arduino Uno, such that the LED turns **ON** automatically in darkness and turns **OFF** automatically in bright light, without any manual switching.

2. COMPONENTS REQUIRED

- Arduino Uno (x1)
- Breadboard (x1)
- LDR – Light Dependent Resistor (x1)
- LED – any colour (x1)
- 220Ω Resistor (current-limiting, for LED)
- 10kΩ Resistor (pull-down, for LDR divider)
- Connecting / jumper wires
- USB cable (power & programming)

3. CIRCUIT CONNECTIONS

1. Connect the LED's positive leg (anode) to Arduino digital pin 13.
2. Connect the LED's negative leg (cathode) through the 220Ω resistor to GND.
3. Connect one leg of the LDR to the 5V supply rail.
4. Connect the other leg of the LDR to analog pin A0.
5. Connect the 10kΩ resistor between pin A0 and GND — this forms a voltage divider with the LDR.

Why a voltage divider? The LDR alone cannot produce a voltage Arduino can read — it only changes resistance. Pairing it with a fixed 10kΩ resistor creates a junction whose voltage rises and falls predictably with light level, which A0 can then measure as an analog value (0-1023).

4. ARDUINO CODE

```
int ldr = A0;
int led = 13;

void setup()
{
  pinMode(led, OUTPUT);
}

void loop()
{
  int value = analogRead(ldr);

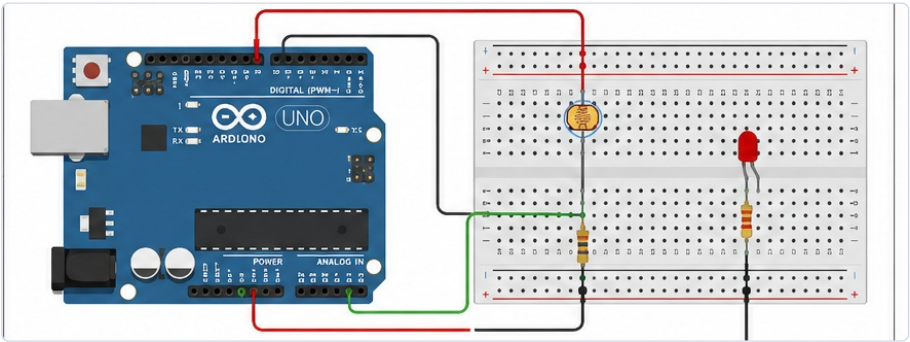
  if(value < 500)
    digitalWrite(led, HIGH); // LED ON in darkness
  else
    digitalWrite(led, LOW);  // LED OFF in bright light
}
```

4a. CODE EXPLANATION (LINE-BY-LINE)

Code Line	Explanation
int ldr = A0;	Creates a variable named <code>ldr</code> and stores the value <code>A0</code> in it. From now on, writing <code>ldr</code> in the code means “analog pin A0” — the pin where the LDR’s voltage-divider output is connected.
int led = 13;	Creates a variable named <code>led</code> storing the value <code>13</code> , the digital pin the LED is wired to. Using named variables instead of raw pin numbers makes the code easier to read and edit later.
void setup() { ... }	A special function that runs only once , right when the Arduino powers on or resets. It is used for one-time configuration.
pinMode(led, OUTPUT);	Configures pin 13 as an OUTPUT pin, telling the Arduino it will be used to send power out (to drive the LED) rather than read a signal in.
void loop() { ... }	A special function that runs repeatedly, forever , right after <code>setup()</code> finishes. All the sensing-and-switching logic lives here so it keeps checking the light level continuously.
int value = analogRead(ldr);	<code>analogRead()</code> measures the voltage at pin A0 and converts it into a number between 0 and 1023 (0 = 0V, 1023 = 5V). This number is stored in a new variable called <code>value</code> and represents the current light intensity.
if(value < 500)	Checks whether the light-intensity reading is below the threshold of 500. A low value here corresponds to darkness , because the LDR’s resistance — and so the divider voltage at A0 — behaves that way in this circuit.
digitalWrite(led, HIGH);	Sends 5V to pin 13, switching the LED ON . This line only runs when the <code>if</code> condition above is true (i.e. it’s dark).
else digitalWrite(led, LOW);	If the condition was false (light level ≥ 500, i.e. bright), this line runs instead, sending 0V to pin 13 and switching the LED OFF .

Why no delay() is needed: Since `loop()` re-executes continuously on its own, the Arduino is already re-checking the LDR value many times per second. This makes the street light respond to changing light conditions in near real-time.

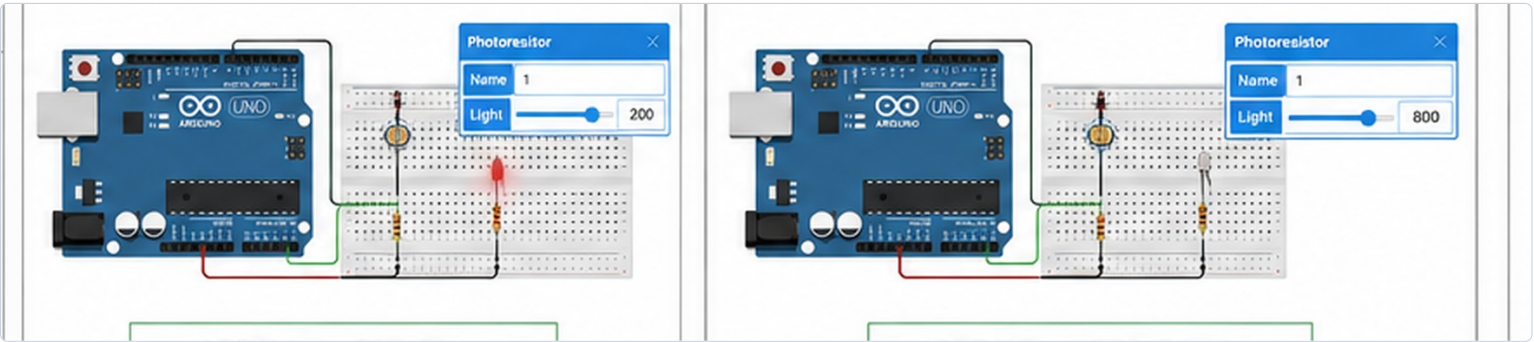
5. CIRCUIT DIAGRAM (TINKERCAD)



- The LDR and the 10kΩ resistor together form a voltage divider.
- Pin A0 reads the resulting analog voltage as a light-intensity value (0–1023).
- The LED, connected to pin 13, switches ON or OFF based on that value.

Wire	Connects to
Red	5V
Black	GND
Green	A0
Blue	Pin 13

6. SIMULATION OUTPUTS



A. LED ON — Dark Condition

Photoresistor light level set to 200 in Tinkercad.

LDR Value = 200 (Dark) → LED is **ON**

B. LED OFF — Bright Condition

Photoresistor light level set to 800 in Tinkercad.

LDR Value = 800 (Bright) → LED is **OFF**

7. WORKING

The LDR senses the intensity of incident light. In darkness, the resistance of the LDR increases, which pulls the voltage — and hence the value read at A0 — **low (< 500)**. The Arduino detects this and turns the LED **ON**. In bright light, the LDR’s resistance decreases, so the value read at A0 becomes **high (> 500)**, and the Arduino turns the LED **OFF**. This closed loop of sensing and switching repeats continuously inside `loop()`, giving the system automatic, real-time response to ambient light — no human intervention needed.

8. OUTPUT TABLE

Light Condition	LDR Value (Approx.)	LED Status
Bright Light	> 500	OFF
Dim Light	400 – 500	May vary (threshold zone)
Darkness	< 500	ON

9. RESULT

The automatic street light using an LDR sensor was successfully designed and simulated. The LED turns ON in darkness and OFF in bright light, exactly as intended. This low-cost system demonstrates a practical, real-life approach to saving electricity by lighting streets only when actually needed.

10. APPLICATIONS

- Automatic street lighting and highway lighting systems.
- Garden, garage, and outdoor security lights.
- Automatic night lamps for homes and corridors.
- Solar-powered lighting systems with light-based switching.
- Automatic display/signboard backlighting.

11. ADVANTAGES & LIMITATIONS

Advantages

- Fully automatic — no manual switching required.
- Saves electricity by lighting up only when dark.
- Simple, low-cost components.
- Easy to scale up with relays for higher-power lights.

Limitations

- LDR response can be affected by dust, shadows, or nearby artificial light.
- Fixed threshold (500) may need calibration for different environments.
- Not weather-sealed in a real outdoor deployment without an enclosure.

12. PRECAUTIONS

- Always connect the LED through the current-limiting resistor (220Ω) to avoid burnout.
- Double-check polarity of the LED (longer leg = anode = positive) before powering the circuit.
- Ensure the LDR voltage-divider connection (with the 10kΩ resistor) is correct, or A0 will read an unstable value.
- Avoid loose breadboard connections; verify continuity before running the simulation.

13. VIVA / QUICK-REVISION QUESTIONS

Q1. What is an LDR?

A Light Dependent Resistor — a component whose resistance decreases as light intensity increases (and increases in darkness).

Q2. Why is a 10kΩ resistor used with the LDR?

To form a voltage divider so the changing resistance of the LDR can be converted into a measurable analog voltage at pin A0.

Q3. What does `analogRead()` return?

An integer between 0 and 1023, proportional to the voltage (0–5V) sensed at the analog pin.

Q4. Why is 500 chosen as the threshold in the code?

It is a calibration midpoint that worked for this divider combination — it can be tuned higher or lower depending on the specific LDR and ambient lighting.

Q5. What real-world advantage does this project demonstrate?

Automatic, sensor-driven electricity saving — lights stay off during the day and switch on only when genuinely needed at night.